Plants have no need of the neurone, because they get their living without moving around, but it is found in the great majority of animal groups. It may have been 'discovered' early in animal evolution, and inherited by all groups, or it may have been rediscovered several times independently.

Neurones are basically just cells, with a nucleus and chromosomes like other cells. But their cell walls are drawn out in long, thin, wire-like projections. Often a neurone has one particularly long 'wire' called the axon. Although the width of an axon is microscopic, its length may be many feet: there are single axons which run the whole length of a giraffe's neck. The axons are usually bundled together in thick multi-stranded cables called nerves. These lead from one part of the body to another carrying messages, rather like trunk telephone cables. Other neurones have short axons, and are confined to dense concentrations of nervous tissue called ganglia, or, when they are very large, brains. Brains may be regarded as analogous in function to computers.* They are analogous in that both types of machine generate complex patterns of output, after analysis of complex patterns of input, and after reference to stored information.

The main way in which brains actually contribute to the success of survival machines is by controlling and coordinating the contractions of muscles. To do this they need cables leading to the muscles, and these are called motor nerves. But this leads to efficient preservation of genes only if the timing of muscle contractions bears some relation to the timing of events in the outside world. It is important to contract the jaw muscles only when the jaws contain something worth biting, and to contract the leg muscles in running patterns only when there is something worth running towards or away from. For this reason, natural selection favoured animals that became equipped with sense organs, devices which translate patterns of physical events in the outside world into the pulse code of the neurones. The brain is connected to the sense organs—eyes, ears, taste-buds, etc.—by means of cables called sensory nerves. The workings of the sensory systems are particularly baffling, because they can achieve far more sophisticated feats of pattern-recognition than the best and most expensive man-made machines; if this were not so, all typists would be redundant, superseded by speech-recognizing machines, or machines for reading handwriting. Human typists will be needed for many decades yet.